Project 1: The 3D Character Trio

Model 3 characters. Add color. Animate 3 parts. Export your video.

Class Time: 40 mins per week	**Software:** Blender (Free 3D Software)
Assessed On: Criteria B (Ideas), C (Making), D (Testing)	**Final Goal:** 5-Second MP4 Animated Looping Video and .blend file.

1. What Are You Making?

You are a 3D toy and character designer! You must make a set of 3 cute, expressive vinyl characters in Blender. Your characters cannot talk, so their shapes and movements must show their emotions.

- **3 Different Shapes:** Character 1 must be a Ball/Sphere; Character 2 must be a Tall Cylinder/Capsule; Character 3 must be a Box/Cuboid with round corners.
- **Clean Colors:** Give each character its own bright, matte toy color (not plain grey).
- **3 Moving Parts:** At least 3 parts in the scene must animate using keyframes (Example: Eyes blink, mouth moves, body hops or squashes).
- **Final Video:** Render a clean 5-second video file (.mp4) that loops smoothly.

2. The Big Idea (Enhanced MYP Context)

Me (Individual)	*How can I make simple 3D shapes look happy, sleepy, or grumpy?*
Our School (Local)	*Can my classmates understand my characters' feelings without hearing words?*
The World (Global)	*Why do simple characters like Fall Guys, Roblox, or Funko connect with kids everywhere in the world?*

3. What You Hand In (Student Deliverables)

Criterion	Deliverable	What Must Be Inside
Criterion B (Ideas)	**Design Sheet (Paper or Slides)**	• 2 different drawing ideas for the trio • Labels showing shapes, colors, and the 3 moving parts • 1 sentence explaining why you picked your final design
Criterion C (Making)	**Blender File & Video (.blend + .mp4)**	• 3 distinct objects modeled (Sphere, Cylinder, Box) • Colored materials applied to all 3 • 3 parts animated with keyframes • 1 paragraph listing any changes you had to make
Criterion D (Evaluating)	**Testing Sheet (1 Page)**	• Test results: 3 classmates rate your animation (1 to 5 stars) • Check your video against your plan: Did everything work? • 2 things you would improve if you had more time

4. Project Grading Rubrics (Criteria B, C, D)

Look at the right-hand column ('What You Hand In') to see exactly what you need to submit for each score.

Criterion B: Developing Ideas (Max Score: 8)

MYP Strands: B.i: Design specification | B.ii: Range of feasible ideas | B.iii: Chosen design & reasons | B.iv: Planning drawings

Score	Official IB Year 3 Descriptor	What You Hand In
0	The student does not reach a standard described by any of the descriptors.	Nothing handed in.
1–2	The student: i. lists a few basic success criteria ii. presents one design idea iii. creates incomplete planning drawings.	• You only drew 1 character idea. • Missing labels and missing moving parts.
3–4	The student: i. constructs a list of success criteria ii. presents a few feasible design ideas iii. outlines reasons for choosing the design iv. creates planning drawings or lists requirements.	• You drew 2 different trio ideas. • You wrote simple notes on shapes and colors. • You drew a basic plan showing which parts will move.
5–6	The student: i. develops design specifications ii. presents a range of feasible design ideas iii. outlines main reasons for selection iv. develops accurate planning drawings and lists requirements.	• You listed clear rules for your 3 characters and 3 moving parts. • You drew 3 neat ideas with labels. • You explained why your chosen trio is best. • You made front/side drawings so anyone can see your plan.
7–8	The student: i. develops a design specification outlining success criteria ii. presents a range of feasible design ideas with annotation iii. outlines reasons for selection with reference to specification iv. develops accurate planning drawings and outlines requirements.	• Complete rules (exact shapes, color names, frame counts for the 3 moving parts). • Multiple creative drawing ideas with clear labels. • Detailed explanation of why your final choice meets all rules. • Accurate turnaround sheets and an animation plan ready for Blender.

Criterion C: Creating the Solution (Max Score: 8)

MYP Strands: C.i: Logical plan outlining time/resources | C.ii: Technical skills | C.iii: Follows plan to create solution | C.iv: Explains changes made

Score	Official IB Year 3 Descriptor	What You Hand In
0	The student does not reach a standard described by any of the descriptors.	No Blender file or video submitted.
1–2	The student: i. demonstrates minimal technical skills ii. creates the solution, which functions poorly and is incomplete.	• You only made 1 or 2 shapes. • Characters have no color or the animation doesn't play.
3–4	The student: i. outlines each step in a plan ii. demonstrates satisfactory technical skills iii. creates the solution, which partially functions iv. outlines changes made to chosen design.	• You modeled 3 shapes with colors. • Only 1 or 2 parts move, or the animation is stuck. • You noted 1 thing you had to fix.
5–6	The student: i. constructs a plan considering time and resources ii. demonstrates competent technical skills iii. creates the solution, which functions as intended iv. outlines changes made to chosen design and plan.	• All 3 distinct characters (Sphere, Cylinder, Box) are modeled cleanly. • All 3 characters have colors applied. • At least 3 parts animate properly on the timeline. • You wrote a short paragraph explaining what you changed.
7–8	The student: i. constructs a logical plan outlining efficient use of time ii. demonstrates excellent technical skills iii. follows plan to create solution, presented	• High-level Blender skills shown (smooth surfaces, neat modifiers, clean keyframing). • All 3 characters look finished and polished with materials. • All 3 parts animate smoothly and loop cleanly in a

	appropriately iv. explains changes made to chosen design and plan.	rendered MP4. • Clear explanation of all fixes and changes made during modeling.

Criterion D: Evaluating (Max Score: 8)

MYP Strands: D.i: Relevant testing methods generating data | D.ii: Success against design specification | D.iii: Improvements | D.iv: Impact on client/audience

Score	Official IB Year 3 Descriptor	What You Hand In
0	The student does not reach a standard described by any of the descriptors.	No testing or reflection handed in.
1–2	The student: i. describes a testing method ii. states the success of the solution.	• You just wrote 'I watched it and it looked okay.'
3–4	The student: i. describes a relevant testing method generating data ii. outlines success against specification iii. lists ways the solution could be improved iv. outlines impact on client/target audience.	• You showed your video to 2 classmates and asked what they thought. • You checked if your 3 moving parts worked. • You wrote 1 thing you would change next time.
5–6	The student: i. describes relevant testing methods generating data ii. describes success against specification based on testing iii. outlines how the solution could be improved iv. describes impact on client/target audience.	• You used a short survey (rating 1 to 5 stars on shape, color, motion). • You compared your final video directly to your Criterion B rules. • You listed 2 specific improvements for your 3D models or animation. • You described what emotions your classmates felt watching them.
7–8	The student: i. describes detailed and relevant testing methods generating accurate data ii. explains success against specification based on authentic testing iii. describes how solution could be improved iv. describes impact on client/target audience.	• Real test data collected (ratings + written feedback from classmates). • You proved how your animation met every project rule with data evidence. • You gave detailed ideas to improve the lighting, timing, and 3D shapes. • You explained clearly how your non-verbal characters connected with your audience.

5. Quick In-Class Checklist (Check off as you work!)

[ ] I have drawn my 3 character ideas and picked my best trio (Criterion B).
[ ] Object 1: Sphere character modeled with eyes, mouth, and legs in Blender (Criterion C).
[ ] Object 2: Tall capsule character modeled with its own unique face (Criterion C).
[ ] Object 3: Rounded cuboid character modeled to match the group (Criterion C).
[ ] All 3 characters have bright, matte colors applied (Criterion C).
[ ] Animated Part 1 is keyframed (e.g. eyes blinking) (Criterion C).
[ ] Animated Part 2 is keyframed (e.g. mouth moving) (Criterion C).
[ ] Animated Part 3 is keyframed (e.g. body hopping or squashing) (Criterion C).
[ ] I rendered my 5-second .mp4 video file and saved it (Criterion C).
[ ] 3 classmates watched my video and filled out my test sheet (Criterion D).